

An Internship Report
Of
AWS Cloud Virtual Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)



By

Roll Number of Student: 2410030818
Name of Student: PUSHKAR KUMAR
Batch: 2024–2028

CANDIDATE'S DECLARATION

I, PUSHKAR KUMAR, do hereby declare that the internship report titled “AWS Cloud Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: PUSHKAR KUMAR

Roll No.: 2410030818

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the AICTE – EduSkills Virtual Internship Program team, EduSkills Academy, and AWS Academy for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of AWS Cloud. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-

Student Name: PUSHKAR KUMAR

Roll No.: 2410030818

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3. INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



AICTE
अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



NATIONAL
INTERNSHIP
PORTAL



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Pushkar Kumar

IILM University, Greater Noida

has successfully completed the 8-weeks

Cloud Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Curriculum Provided by:

 **aws** academy



Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education



Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 4652a018169a8ff5a0d3
Student ID: STU6997f8d8cc4bc1771567320



GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

Internship Offer Letter



Ref No: C16Y2026M081381018

Internship Offer Letter

Student Name : Pushkar Kumar
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : AWS Cloud
Internship Duration : April 2026 - June 2026
AICTE Student ID : STU6997f8d8cc4bc1771567320

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Cloud Foundations	Cloud concepts, economics, and billing while exploring AWS global infrastructure for foundational cloud knowledge and practical understanding.
Week 2	Cloud Foundations	Cloud security, networking, and content delivery, gaining practical knowledge to build secure and efficient cloud solutions.
Week 3	Cloud Foundations	AWS compute, storage, and database services, along with cloud architecture fundamentals for building scalable solutions.
Week 4	Cloud Foundations	Automatic scaling and monitoring in the cloud to optimize performance and manage resources efficiently.
Week 5	Cloud Architecting	Learn cloud architecture, IAM security, S3 storage, EC2 compute, RDS databases, VPC networking, and VPC peering.
Week 6	Cloud Architecting	Secure access using IAM and KMS, implement monitoring, elasticity, and high availability with EC2 and Route 53.
Week 7	Cloud Architecting	Build decoupled architectures using Amazon SQS and design serverless microservices with AWS Lambda.
Week 8	Cloud Architecting	Data engineering patterns, disaster recovery planning, and AWS Solutions Architect certification preparation.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



Bibek Ranjan
Director, EduSkills

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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 8-week Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program, in the domain of “AWS Cloud”, with curriculum provided by AWS Academy. The internship was carried out from April 2026 to June 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The primary aim of this internship was to build a strong practical foundation in cloud computing concepts and Amazon Web Services (AWS), covering cloud foundations and cloud architecting, and to apply this knowledge to designing secure, scalable, and highly available cloud solutions.

The internship followed a structured, week-wise curriculum combining conceptual learning with hands-on labs, weekly assessments, project documentation, and a Final Assessment Test. On successful completion of all the requirements, including the Final Credential Validation / Internship Grade Point Assessment, a Virtual Internship Certificate with an “Outstanding” grade was awarded by EduSkills Academy under the patronage of AICTE and the National Internship Portal.

4.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO/IEC 20000-1:2018 certified organization working under the theme “Nation Building Through Skills”. EduSkills partners with the All India Council for Technical Education (AICTE) to run the AICTE – EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal. For the AWS Cloud track, EduSkills collaborates with AWS Academy to design an industry-aligned curriculum covering cloud foundations and cloud architecting, delivered through structured weekly modules, assessments, and a final assessment test, that helps engineering students build practical cloud-computing capability.

4.3 Problem Statement

Although a large number of engineering students learn the theoretical foundations of computer networks and distributed systems during their academic coursework, very few get structured, hands-on exposure to designing and deploying solutions on an

industry-standard cloud platform such as Amazon Web Services (AWS). Concepts such as cloud security, elastic compute, scalable storage, and highly available architecture are often understood only at a theoretical level, with little practice in configuring, securing, and monitoring real cloud resources. The problem addressed by this internship was to practically learn and apply the AWS cloud computing workflow – from understanding cloud economics and global infrastructure, to securing and networking cloud resources, to architecting scalable, decoupled, and serverless solutions.

4.4 Project Objectives

- To understand core cloud concepts, economics, and billing, and explore the AWS global infrastructure.
- To learn cloud security, networking, and content delivery for building secure and efficient cloud solutions.
- To gain practical knowledge of AWS compute, storage, and database services along with cloud architecture fundamentals.
- To implement automatic scaling and monitoring for optimizing performance and managing cloud resources.
- To learn cloud architecture principles, IAM security, S3 storage, EC2 compute, RDS databases, and VPC networking and peering.
- To secure access using IAM and KMS, and implement monitoring, elasticity, and high availability using EC2 and Route 53.
- To build decoupled architectures using Amazon SQS and design serverless microservices using AWS Lambda.
- To understand data engineering patterns, disaster recovery planning, and prepare for the AWS Certified Solutions Architect certification.
- To successfully clear the Final Credential Validation / Internship Grade Point Assessment.

4.5 Scope of the Project

The scope of this internship spans the complete AWS cloud computing workflow – starting with cloud foundations such as cloud economics, global infrastructure, security,

networking, and content delivery, moving into compute, storage, and database services, and extending into cloud architecting topics such as IAM security, VPC networking, elasticity, high availability, decoupled architectures, and serverless microservices. It further covers data engineering patterns, disaster recovery planning, and preparation for the AWS Certified Solutions Architect certification. The internship was limited to a virtual, simulation/lab-based learning environment provided by AWS Academy and EduSkills Academy, and did not involve deployment on any live organizational production system.

4.6 Technologies and Tools Used

- Cloud Platform: Amazon Web Services (AWS)
- Compute Services: Amazon EC2, AWS Lambda
- Storage & Database: Amazon S3, Amazon RDS
- Networking & Security: Amazon VPC, VPC Peering, IAM, AWS KMS, Amazon Route 53
- Messaging & Integration: Amazon SQS
- Monitoring & Scaling: Amazon CloudWatch, Auto Scaling
- Learning Platform: AWS Academy curriculum, delivered via EduSkills

4.7 System Architecture

The general architecture followed during the internship for the AWS Cloud learning track can be summarized in the following layers:

- Foundation Layer: Cloud concepts, global infrastructure, economics, and billing form the base understanding of the AWS ecosystem.
- Security & Networking Layer: IAM, AWS KMS, Amazon VPC, and VPC peering secure and connect cloud resources.
- Compute & Storage Layer: EC2, Lambda, S3, and RDS provide the core compute, serverless, and data-persistence capabilities.
- Delivery & Availability Layer: Route 53 and content delivery mechanisms ensure high availability and efficient routing.

- Integration Layer: Amazon SQS decouples application components for scalable, asynchronous processing.
- Monitoring & Scaling Layer: Amazon CloudWatch and Auto Scaling optimize performance and manage resources automatically.
- Assessment Layer: Weekly assessments, project documentation, and a Final Assessment Test track progress and final grading.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on labs, and assessments, culminating in a Final Assessment Test, as summarized in the table below.

Week	Module	Description
1	Cloud Foundations	Cloud concepts, economics, and billing while exploring AWS global infrastructure for foundational cloud knowledge and practical understanding.
2	Cloud Foundations	Cloud security, networking, and content delivery, gaining practical knowledge to build secure and efficient cloud solutions.
3	Cloud Foundations	AWS compute, storage, and database services, along with cloud architecture fundamentals for building scalable solutions.
4	Cloud Foundations	Automatic scaling and monitoring in the cloud to optimize performance and manage resources efficiently.
5	Cloud Architecting	Learn cloud architecture, IAM security, S3 storage, EC2 compute, RDS databases, VPC networking, and VPC peering.
6	Cloud Architecting	Secure access using IAM and KMS, implement monitoring, elasticity, and high availability with EC2 and Route 53.
7	Cloud Architecting	Build decoupled architectures using Amazon SQS and design serverless microservices with AWS Lambda.
8	Cloud Architecting	Data engineering patterns, disaster recovery planning, and AWS Solutions Architect certification preparation.
Final	Final Credential Validation	Final Credential Validation / Internship Grade Point Assessment.

Note: There was no stipend associated with this internship.

Each week concluded with weekly assessments and, for certain modules, submission of project documentation and assignments, to evaluate conceptual and practical understanding. In the final week, a Final Assessment Test was conducted, and the overall grade was based on performance across the weekly assessments, project submissions, and the final test.

4.9 Expected Outcomes

- Ability to explain core cloud concepts, economics, and the AWS global infrastructure.
- Practical understanding of cloud security, networking, and content delivery.

- Hands-on exposure to AWS compute, storage, and database services.
- Capability to design scalable and highly available cloud architectures using IAM, S3, EC2, RDS, and VPC.
- Skill to secure access using IAM and KMS, and to implement monitoring and elasticity.
- Ability to build decoupled, serverless architectures using Amazon SQS and AWS Lambda.
- Awareness of data engineering patterns and disaster recovery planning, along with preparation for the AWS Solutions Architect certification.
- Successful clearance of the Final Credential Validation / Internship Grade Point Assessment, resulting in an “Outstanding” grade.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter (Ref No. C16Y2026M081381018) and the Certificate of Virtual Internship issued by EduSkills Academy (Certificate ID: 4652a018169a8ff5a0d3, AICTE Student ID: STU6997f8d8cc4bc1771567320), confirming successful completion of the 8-week AICTE – EduSkills Virtual Internship Program in AWS Cloud with an “Outstanding” grade, are attached in Chapter 3 of this report as proof of completion and communication.

5. BIBLIOGRAPHY/REFERENCES

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